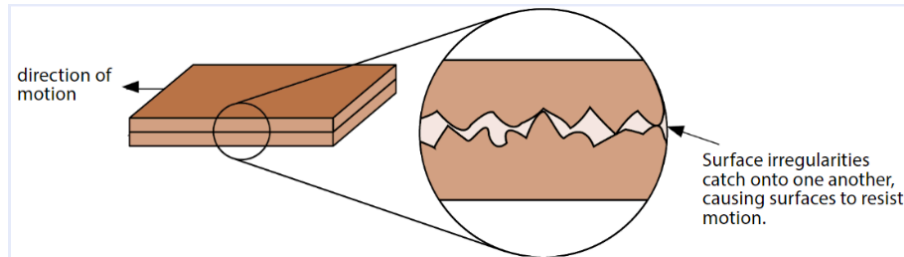


Friction and its Effects

Definition: Friction is the contact force that opposes or tends to oppose motion between surfaces in contact.

#PhysicsDefinitions

⇒ result of irregularities of the surfaces.



Eg; if force is applied to move a book across table towards the right ⇒ friction will act towards the left.

if applied force is removed ⇒ friction will cause the book to slow down and come to a stop eventually.

Positive effects of friction	Negative effects of friction
We can walk without slipping	Cars are less efficient by up to 20%
Moving objects are able to slow down when needed	Moving parts in engines, motors and machine suffer wear and tear

Reducing Negative Effects of Friction

Using Wheels Being circular in shape, wheels greatly reduce the friction between objects and the floor. Smaller force can be applied to move the object around.	Using Ball Bearings Being spherical in shape - ball bearings are used to reduce friction between moving parts of machines, cars, and in line skates. Ball bearings - placed between moving parts so that ball bearings can roll around. Prevents the moving parts from rubbing against each other.
Using Lubricants and polished surfaces Applying a layer of lubricant, such as oil or grease - between surfaces in contact can greatly reduce friction Lubricants - frequently used between moving parts of an engine to reduce wear and tear. Helps to prolong the life of the engine. Polishing a surface removed surface irregularities. Can also reduce friction between surfaces in contact	Using air cushions Friction between two surfaces - reduced - when there is a thin layer of cushion between the 2 surfaces.

Enhancing Positive Effects of Friction

Using Threads	Using Parachute	Using Chalk
Friction - important to motion of vehicles. Without friction → vehicle cannot move as its tires will just spin at the same spot Friction - enables tires to grip the roads surface and roll without slipping Rainy day - moving vehicles may skid on wet roads. Tires need to have more grip on the road to prevent skidding. Thus, tires are designed with threads - grooves that quickly channel water out from underneath the tires. This improves grip of tires on wet roads, thus preventing skidding.	Air resistance (AR) → type of friction in the air. A skydiver in midair varies air resistance to change his speed. - To speed up → he reduces AR by reducing SA in contact with air (head first position) - To slow down → he increases AR by increasing SA in contact with air (spread-eagle position) - To achieve a safe landing - he increases AR significantly → making use of much larger SA of an open parachute.	Rock climbers - need to have firm grip on rock surface with their hands and feet. - Use chalk powder on their hands to absorb perspiration and improve their grip
